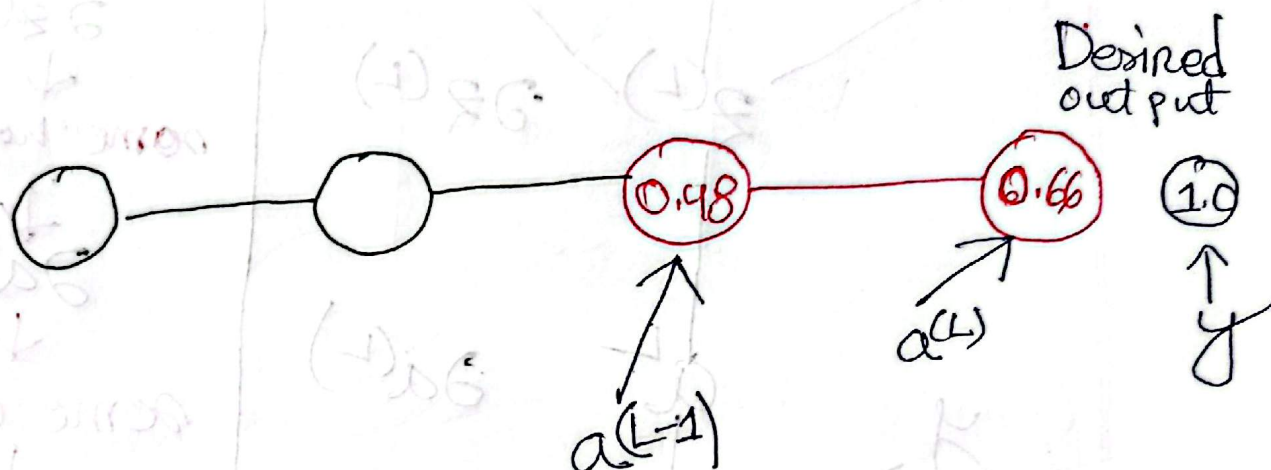


Back Propagation Tutorial :

[Calculus \(3Blue1Brown\)](#)

[Backpropagation, intuitively](#)

Back Propagation



Now,

$$\text{Cost}, C_0 = (a^{(L)} - y)^2$$

$$= (0.66 - 1.0)^2$$

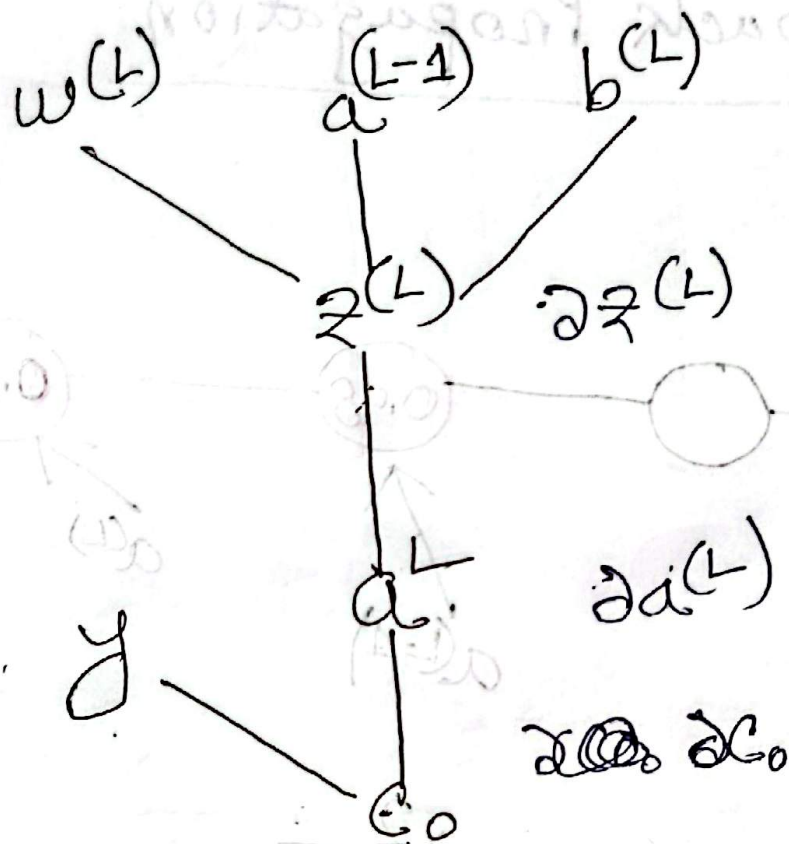
activation of last layer

$$a^{(L)} = \sigma(w^{(L)} a^{(L-1)} + b^{(L)})$$

$$\text{Let, } z^{(L)} = w^{(L)} a^{(L-1)} + b^{(L)}$$

$$\text{and } a^{(L)} = \sigma(z^{(L)})$$

$\partial w^{(L)}$



$\partial w^{(L)}$
 $\downarrow$
 some change
 $\downarrow$
 $\partial z^{(L)}$
 $\downarrow$
 some change
 $\downarrow$
 $\partial a^{(L)}$
 $\downarrow$
 some change
 $\downarrow$
 ∂C_0

$$\frac{\partial C_0}{\partial w^{(L)}} = \frac{\partial z^{(L)}}{\partial w^{(L)}} \cdot \frac{\partial a^{(L)}}{\partial z^{(L)}} \cdot \frac{\partial C_0}{\partial a^{(L)}}$$

$\rightarrow$ give us the sensitivity of C to small changes in $w^{(L)}$

Now, calculate the derivative:

we know,

$$\text{cost} \leftarrow C_0 = (\underbrace{a^{(L)} - y}_{\text{activation of last layer}})^2 \rightarrow \text{True value.}$$

$$\star \frac{\partial C_0}{\partial a^{(L)}} = \frac{\partial}{\partial a^{(L)}} (a^{(L)} - y)^2$$

$$= 2(a^{(L)} - y)$$

$$\star \frac{\partial a^{(L)}}{\partial z^{(L)}} = \frac{\partial}{\partial z^{(L)}} \sigma(z^{(L)})$$

$$= \sigma'(z^{(L)})$$

$$\star \frac{\partial z^{(L)}}{\partial \omega^{(L)}} = \frac{\partial}{\partial \omega^{(L)}} (\omega^{(L)} a^{(L-1)} + b^{(L)})$$

$$= a^{(L-1)}$$

$$\text{So, } \frac{\partial C_0}{\partial \omega^{(L)}} = a^{(L-1)} \sigma'(z^{(L)}) 2(a^{(L)} - y)$$

$$\boxed{\frac{\partial C}{\partial \omega^{(L)}}} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{\partial C_k}{\partial \omega^{(L)}}$$

For one single weight → average of all training example

$$\text{A } \nabla C = \begin{bmatrix} \frac{\partial C}{\partial w^{(1)}} \\ \frac{\partial C}{\partial b^{(1)}} \\ \frac{\partial C}{\partial w^{(L)}} \\ \frac{\partial C}{\partial b^{(L)}} \end{bmatrix}$$

→ ~~For~~ All weight and biases.

Same For,

$$\frac{\partial C_0}{\partial b^{(L)}} = \frac{\partial z^{(L)}}{\partial b^{(L)}} \cdot \frac{\partial a^{(L)}}{\partial z^{(L)}} \cdot \frac{\partial C_0}{\partial a^{(L)}}$$

$$= 1 \cdot \sigma'(z^{(L)}) \cdot (z^{(L)} - y)$$

Note we can not change previous Layer activation directly.